

Database Description

product

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	UUID	Yes	Unique product ID
2			title	VARCHAR(255)	Yes	Product title
3			height	NUMERIC(10,2)	Yes	Product height
4			width	NUMERIC(10,2)	Yes	Product width
5			length	NUMERIC(10,2)	Yes	Product length
6			weight	NUMERIC(10,3)	Yes	Product weight
7			category	VARCHAR(50)	Yes	Product category
8			description	TEXT	No	General product description
9			barcode	VARCHAR(32)	Yes	Unique product barcode
10			original_value	BIGINT	Yes	Original value
11			current_price	BIGINT	Yes	Current selling price
12			stock_quantity	INT	Yes	Current quantity in stock
13			status	VARCHAR(20)	Yes	Product status
14			image_url	VARCHAR(2048)	No	Product image URL
15			created_at	TIMESTAMPTZ	Yes	Product creation time
16			updated_at	TIMESTAMPTZ	Yes	Product update time
17			version	BIGINT	Yes	Product version (for optimistic locking purpose)

book

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	id	UUID	Yes	ID, same as Product ID
2			language	VARCHAR(50)	No	Book language
3			genre	VARCHAR(50)	No	Book genre
4			publication_date	DATE	Yes	Book publication date
5			cover_type	VARCHAR(9)	Yes	Cover type, e.g., paperback/hardcover

6			publisher	VARCHAR(255)	Yes	Book publisher
7			number_of_pages	INTEGER	No	Number of pages

book_authors

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	book_id	UUID	Yes	ID of the book
2	x		author	VARCHAR(255)	Yes	Author name

newspaper

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	id	UUID	Yes	ID, same as Product ID
2			editor_in_chief	VARCHAR(255)	Yes	Editor-in-chief
3			issue_number	VARCHAR(25)	No	Issue number
4			issn	VARCHAR(9)	No	ISSN code
5			publication_frequency	VARCHAR(25)	No	Publication frequency
6			publisher	VARCHAR(255)	Yes	Newspaper publisher
7			language	VARCHAR(50)	No	Newspaper language
8			publication_date	DATE	Yes	Publication date

newspaper_sections

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	newspaper_id	UUID	Yes	ID of the newspaper
2	x		section	VARCHAR(50)	Yes	Newspaper section

cd

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	id	UUID	Yes	ID, same as Product ID
2			record_label	VARCHAR(255)	Yes	CD record label
3			genre	VARCHAR(50)	Yes	CD genre

4			release_date	DATE	No	CD release date
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cd_artists

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	cd_id	UUID	Yes	ID of the CD
2	x		artist	VARCHAR(255)	Yes	Artist name

cd_tracks

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	cd_id	UUID	Yes	ID of the CD containing this track
2	x		track_number	INTEGER	Yes	The track order number
3			title	VARCHAR(255)	Yes	Track title
4			length	BIGINT	Yes	Track length/duration

dvd

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	id	UUID	Yes	ID, same as Product ID
2			studio	VARCHAR(255)	Yes	DVD studio
3			disc_type	VARCHAR(7)	Yes	Disc type, e.g., Blu-ray
4			language	VARCHAR(50)	Yes	DVD language
5			runtime	BIGINT	Yes	Runtime duration
6			director	VARCHAR(255)	Yes	Director name
7			release_date	DATE	No	DVD release date
8			genre	VARCHAR(50)	No	DVD genre

dvd_subtitles

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	dvd_id	UUID	Yes	ID of the DVD
2	x		subtitle	TEXT	Yes	Subtitle language/content

order

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	UUID	Yes	Unique order ID
2			status	VARCHAR(10)	Yes	Order status
3			created_at	TIMESTAMPTZ	Yes	Order creation time
4			updated_at	TIMESTAMPTZ	Yes	Order update time
5			version	BIGINT	Yes	Order version (for optimistic locking purpose)

order_item

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	UUID	Yes	Surrogate ID of the order item
2		x	order_id	UUID	Yes	ID of the order
3		x	product_id	UUID	No	ID of the ordered product
4			quantity	INTEGER	Yes	Ordered quantity
5			product_name	VARCHAR(255)	Yes	Product name at order time
6			unit_price	BIGINT	Yes	Unit price at order time
7			unit_weight	NUMERIC(10,3)	Yes	Unit weight at order time

delivery_information

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	order_id	UUID	Yes	ID of the related order
2			customer_name	VARCHAR(255)	Yes	Customer name
3			customer_email	VARCHAR(255)	Yes	Customer email
4			phone_number	VARCHAR(10)	Yes	Customer phone number
5			province	VARCHAR(50)	Yes	Customer province/city
6			commune	VARCHAR(50)	Yes	Customer commune/ward

7			address	VARCHAR(255)	Yes	Customer address
8			delivery_method	VARCHAR(255)	Yes	Delivery method

invoice

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	order_id	UUID	Yes	ID of the related order
2			total_price_without_vat	BIGINT	Yes	Total product price before VAT
3			total_price_with_vat	BIGINT	Yes	Total product price after VAT
4			delivery_fee	BIGINT	Yes	Delivery fee
5			total_amount	BIGINT	Yes	Final amount to be paid
6			issued_at	TIMESTAMPTZ	Yes	The issue time of the invoice

payment_transaction

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	UUID	Yes	Unique payment transaction ID
2		x	order_id	UUID	Yes	ID of the related order
3			amount_paid	BIGINT	Yes	Amount paid by customer
4			transaction_method	VARCHAR(10)	Yes	Payment method
5			transaction_content	TEXT	Yes	Payment transaction content
6			transaction_timestamp	TIMESTAMPTZ	Yes	Payment transaction time

user

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	UUID	Yes	Unique user ID
2			username	VARCHAR(255)	Yes	User display name

3			email	VARCHAR(255)	Yes	User email, must be unique
4			hashed_password	TEXT	Yes	Securely hashed password
5			active	BOOLEAN	Yes	Whether user is active or not
6			blocked	BOOLEAN	Yes	Whether user is blocked or not
7			must_change_password	BOOLEAN	Yes	Whether user is required to change password on their next login
5			created_at	TIMESTAMPTZ	Yes	User creation time
6			updated_at	TIMESTAMPTZ	Yes	User update time
7			version	BIGINT	Yes	User version (for optimistic locking purpose)

user_roles

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	user_id	UUID	Yes	ID of the user
2	x		role	VARCHAR(20)	Yes	Role of the user

product_log

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	UUID	Yes	Unique product log ID
2		x	manager_id	UUID	No	Product manager who performed the action
3			manager_username	VARCHAR(255)	Yes	Product manager username
4		x	product_id	UUID	No	Product affected by the action
5			product_title	VARCHAR(255)	Yes	Product title

6			timestamp	TIMESTAMP	Yes	Time of the action
7			action	VARCHAR(20)	Yes	Action type, e.g., add, edit, delete

product_update_log

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	log_id	UUID	Yes	Unique ID product log whose action is "UPDATE"
2	x		field_number	UUID	Yes	The order number of the field
3			field_name	VARCHAR(255)	Yes	Name of the field
4			old_value	TEXT	Yes	Old value of the field
5			new_value	TEXT	Yes	New value of the field

stock_adjust_log

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	UUID	Yes	Unique stock adjustment log ID
2		x	manager_id	UUID	No	Product manager who performed the action
3			manager_username	VARCHAR(255)	Yes	Product manager username
4		x	product_id	UUID	No	Product affected by the action
5			product_title	VARCHAR(255)	Yes	Product title
4			old_stock	INT	Yes	Stock before adjustment
5			new_stock	INT	Yes	Stock after adjustment
6			reason	TEXT	Yes	Reason for stock adjustment
7			timestamp	TIMESTAMP	Yes	Time of adjustment

admin_log

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	UUID	Yes	Unique admin log ID
2		x	admin_id	UUID	No	Administrator who performed the action
3			admin_username	VARCHAR(255)	Yes	Administrator username
4		x	affected_user_id	UUID	No	User affected by the action
5			affected_username	VARCHAR(255)	Yes	Affected username
6			timestamp	TIMESTAMP	Yes	Time of action
7			action	VARCHAR(20)	Yes	Action type, e.g., block, unblock, reset password